

Molecular Reaction Dynamics

Theo-1

School on Kinetics and Dynamics of Chemical Reactions

Pablo Gamallo

Materials Science and Physical Chemistry Department
Universitat de Barcelona

material shared with the permission of Prof. Susana Gómez
Universidad de Salamanca

Molecular Reaction Dynamics

Program

- Reaction Dynamics: Introductory concepts
- Collision theory
- Dynamics of elastic and reactive collisions
- Theoretical Methods in Collision Dynamics
- Mechanism of Reactive Collisions
- Potential Energy Surfaces

References:

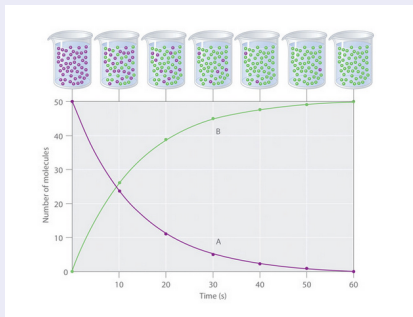
- Chemical Kinetics and Reaction Dynamics, *P.L. Houston*, Dover, 2001.
- Molecular Reaction Dynamics, *R.D. Levine*, Cambridge University Press, 2009.
- Tutorials in Molecular Reaction Dynamics, *M. Brouard and C. Vallance*, RSC Publishing, 2012.

Contents

- 1 Reaction Dynamics: Introductory concepts
- 2 Collision Theory
- 3 Dynamics of elastic and reactive collisions
- 4 Theoretical Methods in Collision Dynamics
- 5 Mechanism of Reactive Collisions
- 6 Potential Energy Surfaces

Molecular Reaction Dynamics

Macroscopic Kinetic Experiments



$$-\frac{dN_A}{dt} \implies K(T) \quad \text{Rate coefficients}$$

Loss of information due to the **thermal averaging**

Molecular Reaction Dynamics

S_N2 Reaction: $\text{Cl}^- + \text{CH}_3\text{Br} \rightarrow \text{ClCH}_3 + \text{Br}^-$ (nucleophilic substitution)

Even very simple reactions display a noticeable complexity when considered at a molecular level.

Molecular Reaction Dynamics

S_N2 Reaction: $\text{Cl}^- + \text{CH}_3\text{Br} \rightarrow \text{ClCH}_3 + \text{Br}^-$ (nucleophilic substitution)

Even very simple reactions display a noticeable complexity when considered at a molecular level.

Reactive collisions

Direct mechanism ▷	Indirect mechanism ($\text{Cl}^- - \text{CH}_3\text{Br}$ complex) ▷	Indirect mechanism ($\text{Cl}^- - \text{CH}_3\text{Br}$ and $\text{ClCH}_3 - \text{Br}^-$ complexes) ▷
$E_{rel} = 50 \text{ kcal/mol}$	$E_{rel} = 1.0 \text{ kcal/mol}$	$E_{rel} = 1.0 \text{ kcal/mol}$

Molecular Reaction Dynamics

S_N2 Reaction: $\text{Cl}^- + \text{CH}_3\text{Br} \rightarrow \text{ClCH}_3 + \text{Br}^-$ (nucleophilic substitution)

Even very simple reactions display a noticeable complexity when considered at a molecular level.

Reactive collisions

Direct mechanism ▷	Indirect mechanism ($\text{Cl}^- - \text{CH}_3\text{Br}$ complex) ▷	Indirect mechanism ($\text{Cl}^- - \text{CH}_3\text{Br}$ and $\text{ClCH}_3 - \text{Br}^-$ complexes) ▷
$E_{\text{rel}} = 50 \text{ kcal/mol}$	$E_{\text{rel}} = 1.0 \text{ kcal/mol}$	$E_{\text{rel}} = 1.0 \text{ kcal/mol}$

Non-reactive collision

Direct mechanism ▷	Indirect mechanism ($\text{Cl}^- - \text{CH}_3\text{Br}$ complex) ▷	Indirect mechanism ($\text{Cl}^- - \text{CH}_3\text{Br}$ and $\text{ClCH}_3 - \text{Br}^-$ complexes) ▷
$E_{\text{rel}} = 1.0 \text{ kcal/mol}$	$E_{\text{rel}} = 1.0 \text{ kcal/mol}$	$E_{\text{rel}} = 1.0 \text{ kcal/mol}$

Molecular Reaction Dynamics

What is Molecular Dynamics?

Molecular Reaction Dynamics

What is Molecular Dynamics?

- It seeks to understand what actually takes place at that level when a change, chemical or physical, occurs.
- It studies elementary processes and the means of **probing**, **understanding** and **controlling** them.

Types of molecular collisions

Non-reactive collisions: $A+BC \rightarrow A+BC$

There is no change in the chemical species.

- Elastic collisions
- Inelastic collisions

Types of molecular collisions

Non-reactive collisions: $A+BC \rightarrow A+BC$

There is no change in the chemical species.

- Elastic collisions
- Inelastic collisions

Reactive collisions: $A+BC \rightarrow AB+C, AC+B, \dots$

The chemical species change.

Non-reactive molecular collisions

Non-reactive molecular collisions

Elastic collisions:



The chemical species, as well as their internal states, do not change as a consequence of the collision.

↪ Translational energy is conserved.

Non-reactive molecular collisions

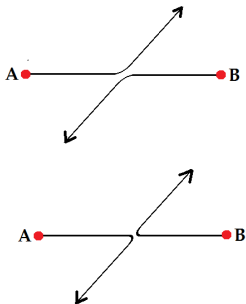
Elastic collisions:



The chemical species, as well as their internal states, do not change as a consequence of the collision.

↪ Translational energy is conserved.

The collision products may be **deflected**, that is, the recoil direction does not necessarily coincides with the incoming direction.



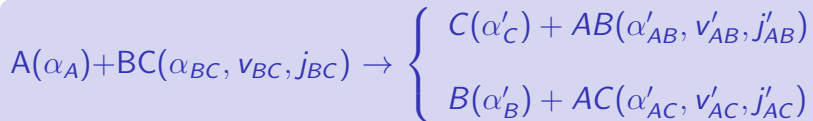
Non-reactive molecular collisions

Inelastic collisions:



The chemical species do not change. However, the internal state of reactants and products is not the same.

Reactive molecular collisions



The chemical species change, along with the internal state of reactants and products.

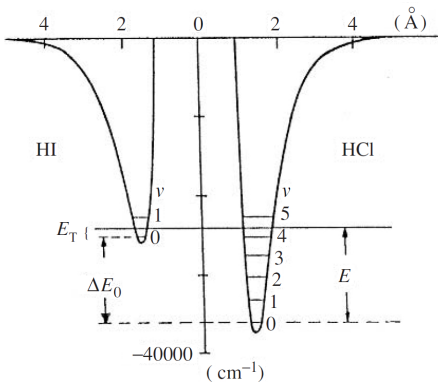
Energy disposal in a chemical reaction $Cl + HI \rightarrow HCl + I$

Energy disposal in a chemical reaction $Cl + HI \rightarrow HCl + I$

Energetics of the reaction $Cl + HI \rightarrow HCl + I$

Energy disposal in a chemical reaction $Cl + HI \rightarrow HCl + I$

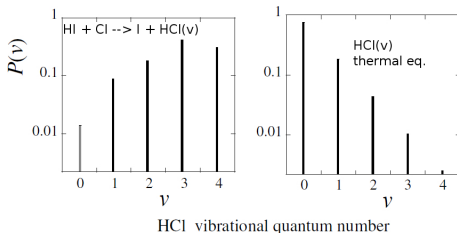
Energetics of the reaction $Cl + HI \rightarrow HCl + I$



- Exoergicity (ΔE_0)
- Translational energy of reactants (E_T)
- Energy disposal (E):
 - ▶ $E = E_T + \Delta E_0$
- Distribution of the energy disposal (E):
 - ▶ vib, rot and trans of products
- Elementary chemical event
 - ↪ isolated system
- Experimental challenge !!
 - ↪ observing individual reactive collisions

Energy disposal in a chemical reaction $Cl + HI \rightarrow HCl + I$

• Distribution of product's energy states



- ▶ Nascent HCl product (**left**):
↪ population inversion
- ▶ Thermal equilibrium (**right**):
↪ Boltzmann equilibrium distribution

Energy disposal in a chemical reaction $Cl + HI \rightarrow HCl + I$

- **Spectator model in a collision**

- ↪ Heavy-Light-Heavy system ($Cl + H - I$)
- ↪ Momenta of I and Cl atoms are conserved
- ↪ Little energy transfer to translation
- ↪ Exoergicity in internal motion of HCl

- **Predictions: products' angular distribution**

- ▶ *Forward* direction
- ▶ *Anisotropic* angular distribution
- ▶ No long-lived intermediate (direct mechanism)

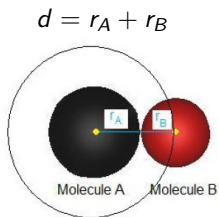
Contents

- 1 Reaction Dynamics: Introductory concepts
- 2 Collision Theory
- 3 Dynamics of elastic and reactive collisions
- 4 Theoretical Methods in Collision Dynamics
- 5 Mechanism of Reactive Collisions
- 6 Potential Energy Surfaces

Collision theory: fundamentals

Collision theory: fundamentals

Rigid spheres model: molecule B approaches molecule A



If the distance between the centers $\leq d$

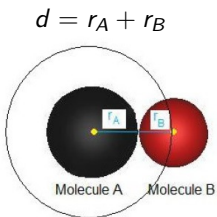
$\Rightarrow$ Interaction

If the distance between the centers $> d$

$\Rightarrow$ No Interaction

Collision theory: fundamentals

Rigid spheres model: molecule B approaches molecule A



If the distance between the centers $\leq d$

$\Rightarrow$ Interaction

If the distance between the centers $> d$

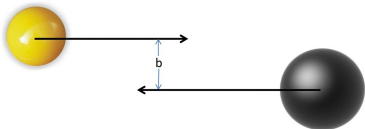
$\Rightarrow$ No Interaction

Collision cross section σ_c

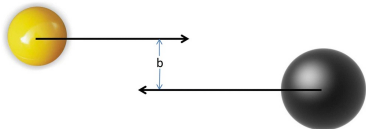
Cross sectional area that the centres of two particles must lie within if they are to collide

$$\sigma_c = \pi d^2$$

Collision theory: fundamentals



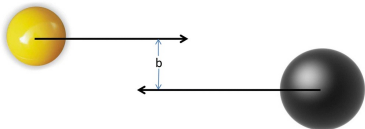
Collision theory: fundamentals



Impact parameter, b

- Quantifies the initial perpendicular separation of the paths of the collision partners
- $b_{\max}$: maximum value of the impact parameter for which a collision can occur $\sigma_c = \pi d^2 \Rightarrow \sigma_c = \pi b_{\max}^2$

Collision theory: fundamentals



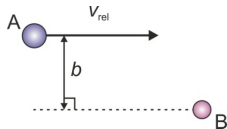
Impact parameter, b

- Quantifies the initial perpendicular separation of the paths of the collision partners
- $b_{\max}$: maximum value of the impact parameter for which a collision can occur $\sigma_c = \pi d^2 \Rightarrow \sigma_c = \pi b_{\max}^2$

Relative velocity, v_{rel}

Average relative velocity between gas molecules A and B

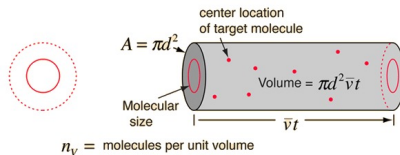
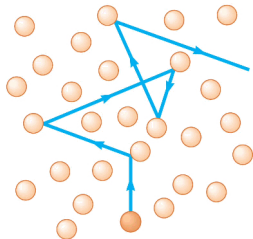
$$\langle v_{rel} \rangle = \left(\frac{8k_B T}{\pi \mu_{AB}} \right)^{\frac{1}{2}}$$



Collision theory: fundamentals

Mean free path, λ

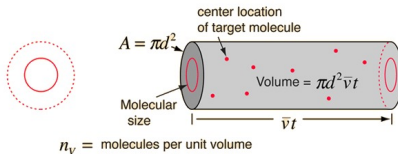
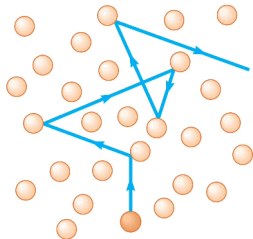
How far, in average, a molecule moves before colliding



Collision theory: fundamentals

Mean free path, λ

How far, in average, a molecule moves before colliding

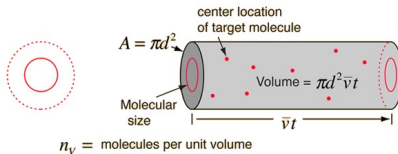
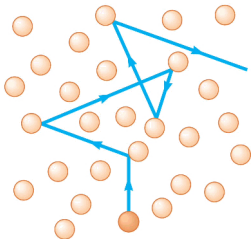


$$\lambda = (n_v \sigma_c)^{-1}$$

Collision theory: fundamentals

Mean free path, λ

How far, in average, a molecule moves before colliding



$$\lambda = (n_v \sigma_c)^{-1}$$

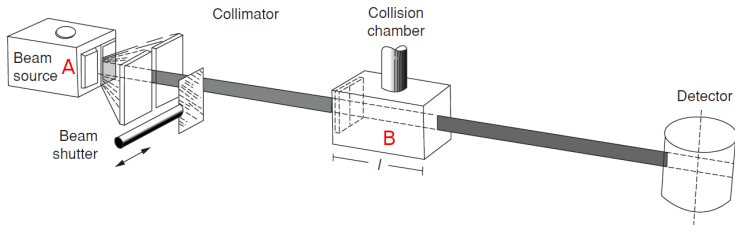
The larger the collision cross section σ_c :

↪ The smaller the mean free path λ

↪ The molecular collisions are more frequent

Collision theory: fundamentals

Direct determination of the mean free path, λ



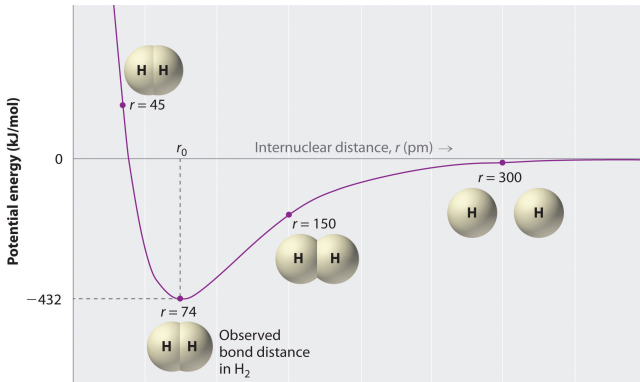
A molecule collides with a target molecule **B**: there is a change in its direction of velocity

$$\ln \frac{I_t}{I_0} = \frac{l}{\lambda}$$

Collision theory: fundamentals

Collision theory: fundamentals

Realistic intermolecular potentials: Of course real molecules are not rigid spheres.



Still we can interpret the collision cross section in the same way, $\lambda = (n_v \sigma_c)^{-1}$.

$\sigma_c \Rightarrow$ intermolecular potentials

Contents

- 1 Reaction Dynamics: Introductory concepts
- 2 Collision Theory
- 3 Dynamics of elastic and reactive collisions**
- 4 Theoretical Methods in Collision Dynamics
- 5 Mechanism of Reactive Collisions
- 6 Potential Energy Surfaces

Dynamics of elastic molecular collisions

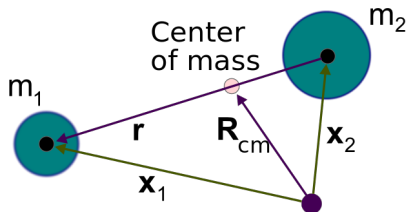
Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.

Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

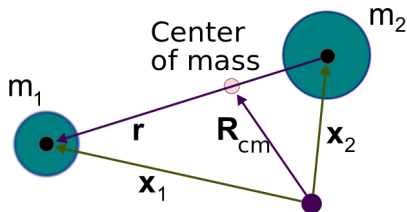
It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.



Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.

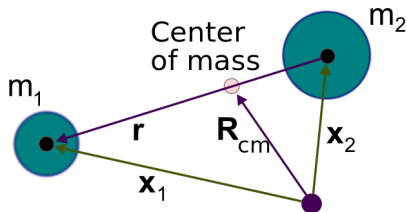


$$\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$$
$$\mathbf{R}_{cm} = \frac{m_1}{M} \mathbf{x}_1 + \frac{m_2}{M} \mathbf{x}_2$$

Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.



$$\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$$

$$\mathbf{R}_{cm} = \frac{m_1}{M} \mathbf{x}_1 + \frac{m_2}{M} \mathbf{x}_2$$

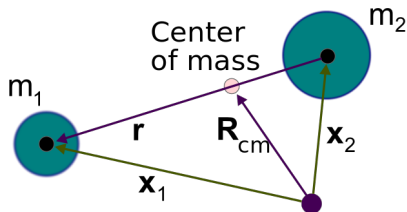
$$\mathbf{v}_i = |\dot{\mathbf{x}}_i|; \mathbf{v}_r = |\dot{\mathbf{r}}|; \mathbf{V}_{cm} = |\dot{\mathbf{R}}_{cm}|$$

$$M = m_1 + m_2 \quad \mu = \frac{m_1 m_2}{M}$$

Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.



$$\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$$

$$\mathbf{R}_{cm} = \frac{m_1}{M} \mathbf{x}_1 + \frac{m_2}{M} \mathbf{x}_2$$

$$\mathbf{v}_i = |\dot{\mathbf{x}}_i|; \mathbf{v}_r = |\dot{\mathbf{r}}|; \mathbf{V}_{cm} = |\dot{\mathbf{R}}_{cm}|$$

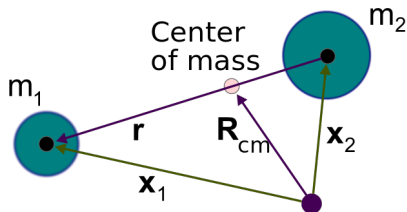
$$M = m_1 + m_2 \quad \mu = \frac{m_1 m_2}{M}$$

$$T = \frac{1}{2} m_1 \mathbf{v}_1^2 + \frac{1}{2} m_2 \mathbf{v}_2^2 =$$

Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.



$$\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$$

$$\mathbf{R}_{cm} = \frac{m_1}{M} \mathbf{x}_1 + \frac{m_2}{M} \mathbf{x}_2$$

$$\mathbf{v}_i = |\dot{\mathbf{x}}_i|; \mathbf{v}_r = |\dot{\mathbf{r}}|; \mathbf{V}_{cm} = |\dot{\mathbf{R}}_{cm}|$$

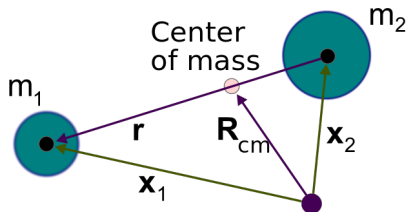
$$M = m_1 + m_2 \quad \mu = \frac{m_1 m_2}{M}$$

$$T = \frac{1}{2} m_1 \mathbf{v}_1^2 + \frac{1}{2} m_2 \mathbf{v}_2^2 = \frac{1}{2} \mu \mathbf{v}_r^2 + \frac{1}{2} M \mathbf{V}_{cm}^2$$

Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.



$$\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$$

$$\mathbf{R}_{cm} = \frac{m_1}{M} \mathbf{x}_1 + \frac{m_2}{M} \mathbf{x}_2$$

$$\mathbf{v}_i = |\dot{\mathbf{x}}_i|; \mathbf{v}_r = |\dot{\mathbf{r}}|; \mathbf{V}_{cm} = |\dot{\mathbf{R}}_{cm}|$$

$$M = m_1 + m_2 \quad \mu = \frac{m_1 m_2}{M}$$

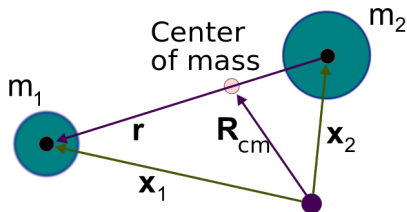
$$T = \frac{1}{2} m_1 \mathbf{v}_1^2 + \frac{1}{2} m_2 \mathbf{v}_2^2 = \frac{1}{2} \mu \mathbf{v}_r^2 + \frac{1}{2} M \mathbf{V}_{cm}^2$$

$$\text{c.o.m frame} \implies \mathbf{V}_{cm} = 0$$

Dynamics of elastic molecular collisions

Coordinate system representation: the center-of-mass frame

It is advantageous to describe a collision in terms of their **relative internal motion** and the **motion of the center-of-mass**.



$$\mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2$$

$$\mathbf{R}_{cm} = \frac{m_1}{M} \mathbf{x}_1 + \frac{m_2}{M} \mathbf{x}_2$$

$$\mathbf{v}_i = |\dot{\mathbf{x}}_i|; \mathbf{v}_r = |\dot{\mathbf{r}}|; \mathbf{V}_{cm} = |\dot{\mathbf{R}}_{cm}|$$

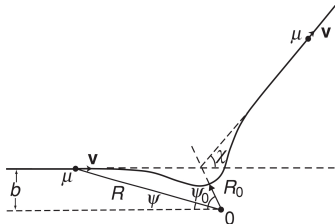
$$M = m_1 + m_2 \quad \mu = \frac{m_1 m_2}{M}$$

$$T = \frac{1}{2} m_1 \mathbf{v}_1^2 + \frac{1}{2} m_2 \mathbf{v}_2^2 = \frac{1}{2} \mu \mathbf{v}_r^2 + \frac{1}{2} M \mathbf{V}_{cm}^2$$

$$\text{c.o.m frame} \Rightarrow \mathbf{V}_{cm} = 0 \Rightarrow T = \frac{1}{2} \mu \mathbf{v}_r^2$$

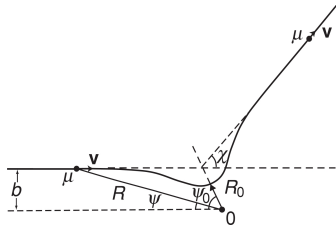
Dynamics of elastic molecular collisions

Dynamics of elastic molecular collisions



- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

Dynamics of elastic molecular collisions

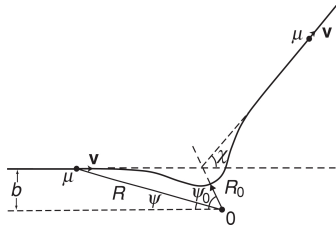


- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

How is a collision trajectory specified?

During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

Dynamics of elastic molecular collisions



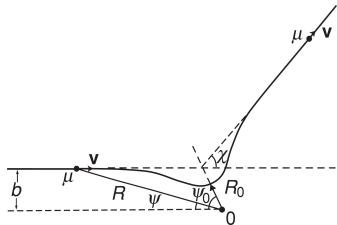
- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

How is a collision trajectory specified?

During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

$$T = \frac{1}{2} \mu \dot{\mathbf{R}}^2$$

Dynamics of elastic molecular collisions



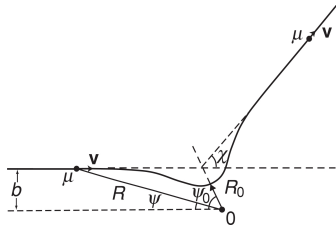
- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

How is a collision trajectory specified?

During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

$$T = \frac{1}{2}\mu\dot{\mathbf{R}}^2 = \frac{1}{2}\mu\left[\dot{R}^2 + R^2\dot{\Psi}^2\right]$$

Dynamics of elastic molecular collisions



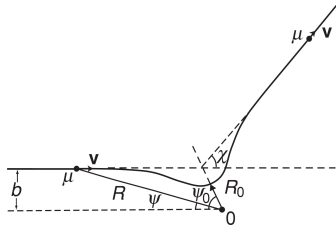
- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

How is a collision trajectory specified?

During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

$$T = \frac{1}{2}\mu\dot{\mathbf{R}}^2 = \frac{1}{2}\mu \left[\dot{R}^2 + R^2\dot{\psi}^2 \right] \text{ and } \dot{\psi} = bv/R^2$$

Dynamics of elastic molecular collisions



- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

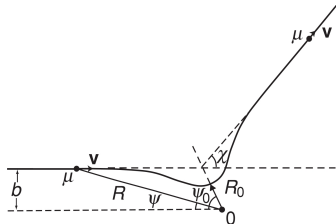
How is a collision trajectory specified?

During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

$$T = \frac{1}{2}\mu\dot{\mathbf{R}}^2 = \frac{1}{2}\mu \left[\dot{R}^2 + R^2\dot{\psi}^2 \right] \text{ and } \dot{\psi} = bv/R^2$$

$$T = \frac{1}{2}\mu \left[\dot{R}^2 + b^2v^2/R^2 \right]$$

Dynamics of elastic molecular collisions



- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

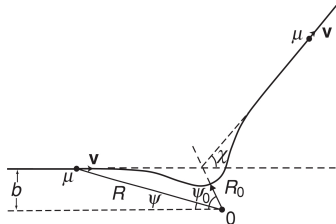
How is a collision trajectory specified?

During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

$$T = \frac{1}{2}\mu\dot{\mathbf{R}}^2 = \frac{1}{2}\mu \left[\dot{R}^2 + R^2\dot{\psi}^2 \right] \text{ and } \dot{\psi} = bv/R^2$$

$$T = \frac{1}{2}\mu \left[\dot{R}^2 + b^2v^2/R^2 \right] \stackrel{E_T = \frac{1}{2}\mu v^2}{=} \frac{1}{2}\mu\dot{R}^2 + E_T b^2/R^2$$

Dynamics of elastic molecular collisions



- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

How is a collision trajectory specified?

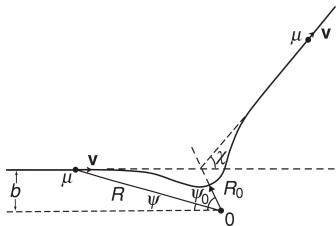
During the trajectory, the relative position vector $\mathbf{R}(t)$ changes with time.

$$T = \frac{1}{2}\mu\dot{\mathbf{R}}^2 = \frac{1}{2}\mu \left[\dot{R}^2 + R^2\dot{\Psi}^2 \right] \text{ and } \dot{\Psi} = bv/R^2$$

$$T = \frac{1}{2}\mu \left[\dot{R}^2 + b^2v^2/R^2 \right] \stackrel{E_T = \frac{1}{2}\mu v^2}{=} \frac{1}{2}\mu\dot{R}^2 + E_T b^2/R^2$$

Important: A unique collision trajectory is defined by specifying the initial collisional energy (E_T) and the impact parameter (b)

Dynamics of elastic molecular collisions

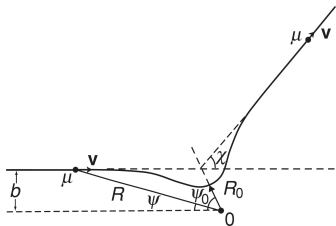


- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

Conservation of angular momentum $\mathbf{L}$

$$\mathbf{L} = \mathbf{R} \times \mathbf{p}$$

Dynamics of elastic molecular collisions

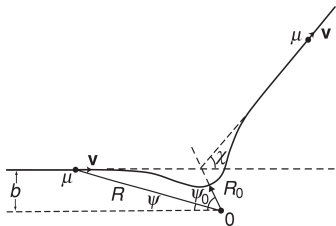


- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

Conservation of angular momentum $\mathbf{L}$

$$\mathbf{L} = \mathbf{R} \times \mathbf{p} \Rightarrow |\mathbf{L}| = |\mathbf{R} \times \mathbf{p}| = |\mu \mathbf{R} \times \mathbf{v}| = \mu v R \sin \Psi$$

Dynamics of elastic molecular collisions



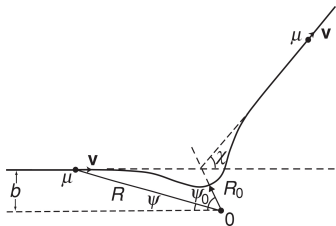
- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

Conservation of angular momentum $\mathbf{L}$

$$\mathbf{L} = \mathbf{R} \times \mathbf{p} \Rightarrow |\mathbf{L}| = |\mathbf{R} \times \mathbf{p}| = |\mu \mathbf{R} \times \mathbf{v}| = \mu v R \sin \Psi$$

- At large separations ($R \rightarrow \infty$): $|\mathbf{L}| = L = \mu v R \sin \Psi =$

Dynamics of elastic molecular collisions



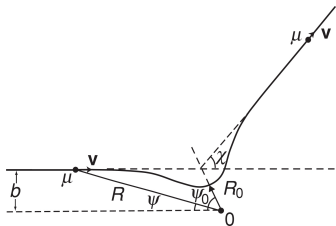
- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

Conservation of angular momentum $\mathbf{L}$

$$\mathbf{L} = \mathbf{R} \times \mathbf{p} \Rightarrow |\mathbf{L}| = |\mathbf{R} \times \mathbf{p}| = |\mu \mathbf{R} \times \mathbf{v}| = \mu v R \sin \Psi$$

- At large separations ($R \rightarrow \infty$): $|\mathbf{L}| = L = \mu v R \sin \Psi = \mu v b$

Dynamics of elastic molecular collisions



- $\mathbf{v}$: initial velocity vector
- b : impact parameter
- $\mathbf{R}$: relative position vector
- Ψ : orientation angle
- χ : deflection angle
- $\mathbf{R}_0$: vector of closest approach

Conservation of angular momentum $\mathbf{L}$

$$\mathbf{L} = \mathbf{R} \times \mathbf{p} \Rightarrow |\mathbf{L}| = |\mathbf{R} \times \mathbf{p}| = |\mu \mathbf{R} \times \mathbf{v}| = \mu v R \sin \Psi$$

- At large separations ($R \rightarrow \infty$): $|\mathbf{L}| = L = \mu v R \sin \Psi = \mu v b$
- $T = \frac{1}{2} \mu \dot{R}^2 + \frac{L^2}{2\mu R^2}$

Dynamics of elastic molecular collisions

Conservation of total energy E

Dynamics of elastic molecular collisions

Conservation of total energy E

When $V(R)=0$ kinetic energy is conserved:

$$E_T = \frac{1}{2}\mu v^2$$

Dynamics of elastic molecular collisions

Conservation of total energy E

When $V(R)=0$ kinetic energy is conserved:

$$E_T = \frac{1}{2}\mu v^2 = \frac{1}{2}\mu \dot{R}^2 + \frac{L^2}{2\mu R^2}$$

Dynamics of elastic molecular collisions

Conservation of total energy E

When $V(R)=0$ kinetic energy is conserved:

$$E_T = \frac{1}{2}\mu v^2 = \frac{1}{2}\mu \dot{R}^2 + \frac{L^2}{2\mu R^2}$$

While when there is a force acting between particles $V(R)\neq 0$:

$$E = T + V(R) = \frac{1}{2}\mu \dot{R}^2 + \frac{L^2}{2\mu R^2} + V(R) = E_T$$

Dynamics of elastic molecular collisions

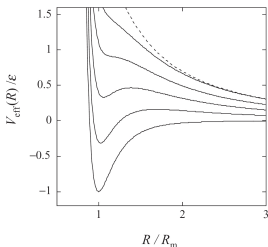
Conservation of total energy E

When $V(R)=0$ kinetic energy is conserved:

$$E_T = \frac{1}{2}\mu v^2 = \frac{1}{2}\mu \dot{R}^2 + \frac{L^2}{2\mu R^2}$$

While when there is a force acting between particles $V(R)\neq 0$:

$$E = T + V(R) = \frac{1}{2}\mu \dot{R}^2 + \frac{L^2}{2\mu R^2} + V(R) = E_T$$

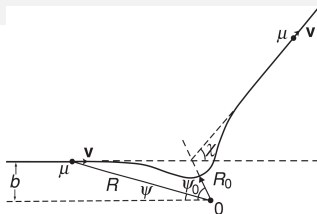


- E_T : Total energy before the collision
- **Effective potential** ($L = \mu vb$):

$$V_{\text{eff}}(R) = \frac{L^2}{2\mu R^2} + V(R)$$

- **Centrifugal barrier**: repulsive contribution to $V_{\text{eff}}(R)$

Deflection angle



Deflection χ : Angle between the initial and final relative velocity

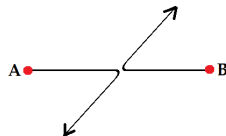
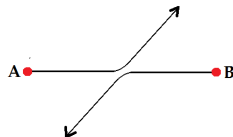
Relation between impact parameter b and deflection angle χ :

- High $b \rightarrow$ hardly collision $\rightarrow$ forward scattering

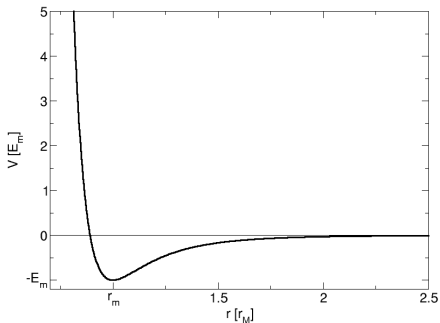
$$b \rightarrow \infty \Rightarrow \chi \rightarrow 0$$

- Low $b \rightarrow$ head-on collision $\rightarrow$ backward scattering

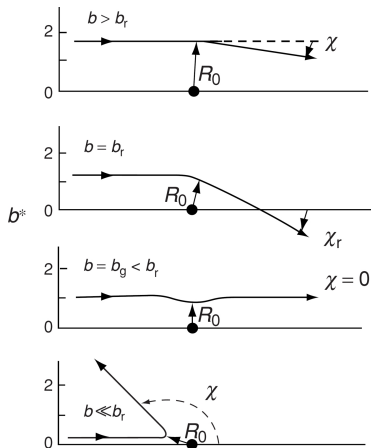
$$b \rightarrow 0 \Rightarrow \chi \rightarrow \pi$$



Deflection angle



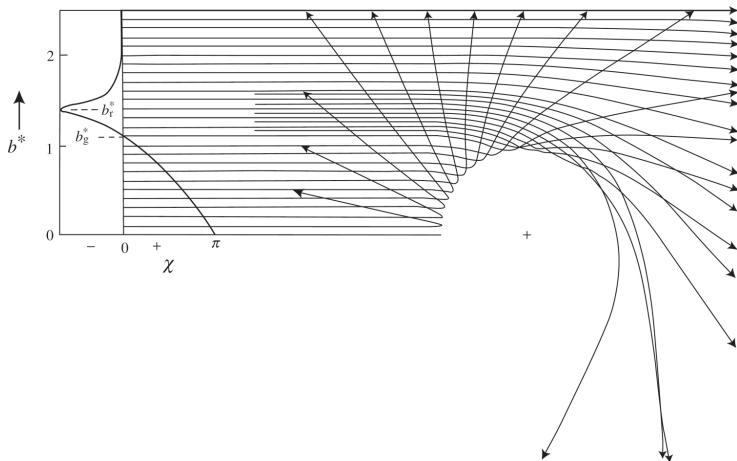
- $b > b_r$: long-range force
- $b = b_r \approx R_m$: maximum deflection (χ_r)
- $b < b_r$: appears repulsive potential
- $b = b_g$: attraction = repulsion ($\chi = 0$)
- $b < b_g$: repulsion dominates



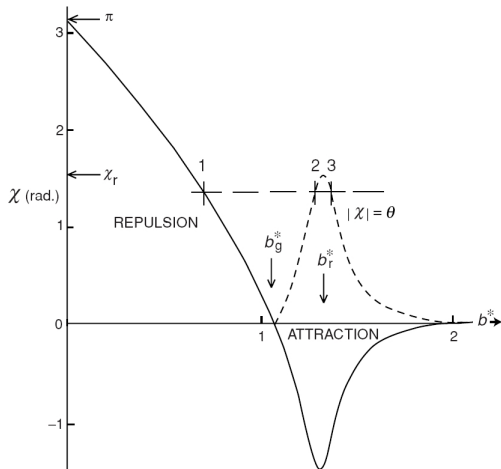
$$b^* = b/r_m$$

R_0 : Distance of closest approach

Deflection function, $\chi(b)$



Deflection function, $\chi(b)$



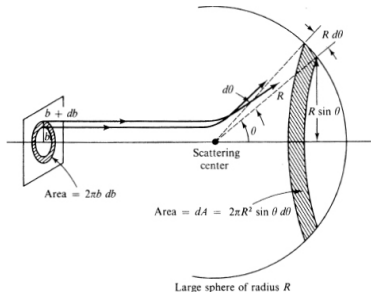
Because of the cylindrical symmetry of the collision process, **experimentally is not possible to distinguish between positive and negative deflection angles.**

$$\theta = |\chi|$$

Dynamics of elastic molecular collisions

The total collision cross-section σ :

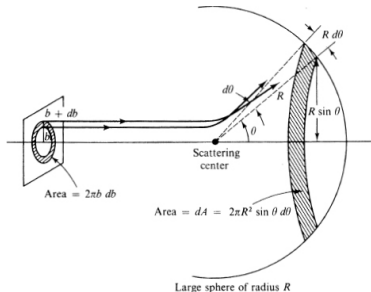
Let us consider collision trajectories with an impact parameter in the range b to $b + db$



Dynamics of elastic molecular collisions

The total collision cross-section σ :

Let us consider collision trajectories with an impact parameter in the range b to $b + db$

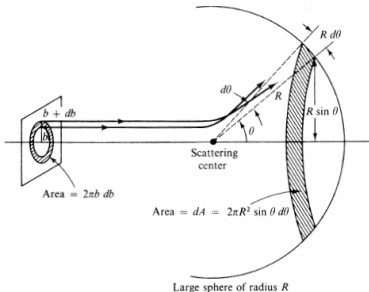


$$\bullet \sigma = \pi b^2$$

Dynamics of elastic molecular collisions

The total collision cross-section σ :

Let us consider collision trajectories with an impact parameter in the range b to $b + db$

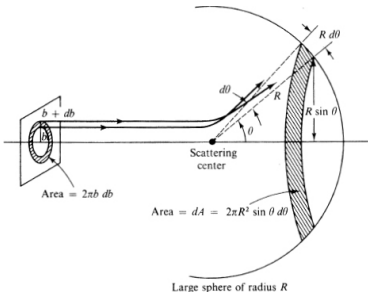


$$\bullet \quad \sigma = \pi b^2 \implies d\sigma = 2\pi b \, db$$

Dynamics of elastic molecular collisions

The total collision cross-section σ :

Let us consider collision trajectories with an impact parameter in the range b to $b + db$



- $\sigma = \pi b^2 \implies d\sigma = 2\pi b db$

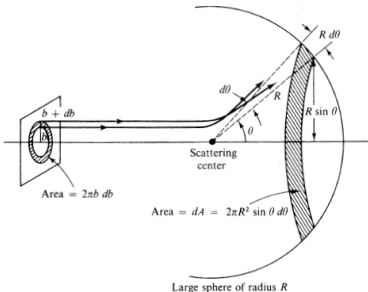
- The total cross-section will be:

$$\sigma = \int d\sigma = \int 2\pi b db$$

Dynamics of elastic molecular collisions

The total collision cross-section σ :

Let us consider collision trajectories with an impact parameter in the range b to $b + db$



- $\sigma = \pi b^2 \implies d\sigma = 2\pi b db$

- The total cross-section will be:

$$\sigma = \int d\sigma = \int 2\pi b db$$

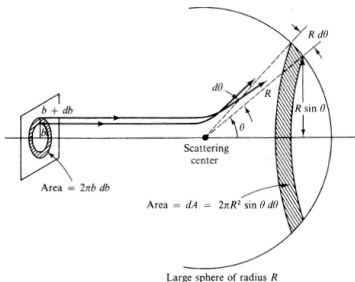
- For example, for rigid spheres:

$$\sigma = \int_0^d 2\pi b db = \pi d^2$$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

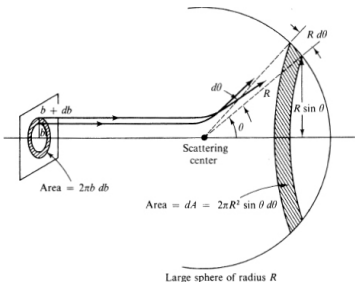
Probability of scattering into a given solid angle $d\Omega = \sin \theta d\theta d\phi$, where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.



Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin \theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

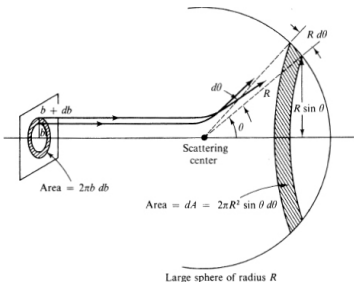


• **central potential** $\Rightarrow$ No ϕ dependence

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin\theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

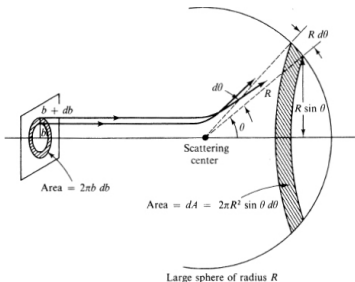


- **central potential** $\implies$ No ϕ dependence
- $d\Omega = \sin\theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin\theta d\theta$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin \theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

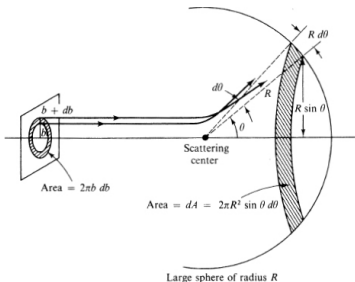


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin \theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin \theta d\theta$
- $d\sigma = 2\pi b db$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin\theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

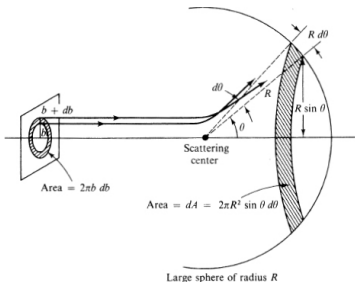


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin\theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin\theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin\theta d\theta}$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin\theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

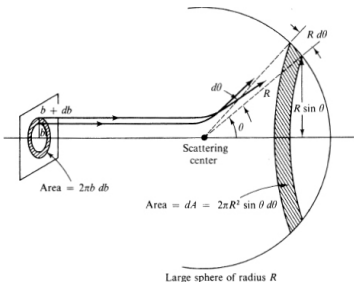


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin\theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin\theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin\theta d\theta} = \frac{b}{\sin\theta (d\theta/db)}$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin \theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

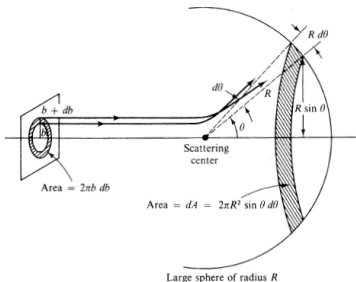


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin \theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin \theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin \theta d\theta} = \frac{b}{\sin \theta (d\theta/db)} = \frac{b}{\sin \theta |d\chi/db|}$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin \theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

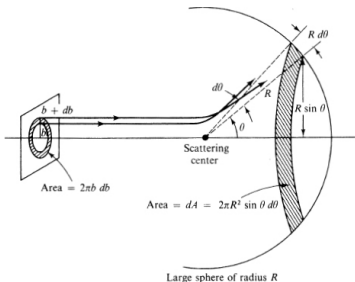


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin \theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin \theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin \theta d\theta} = \frac{b}{\sin \theta (d\theta/db)} = \frac{b}{\sin \theta |d\chi/db|} = I(\theta)$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin\theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

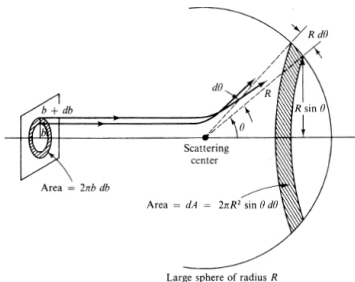


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin\theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin\theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin\theta d\theta} = \frac{b}{\sin\theta (d\theta/db)} = \frac{b}{\sin\theta |d\chi/db|} = I(\theta)$
- $\sigma = \int \frac{d\sigma}{d\Omega} d\Omega$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin \theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

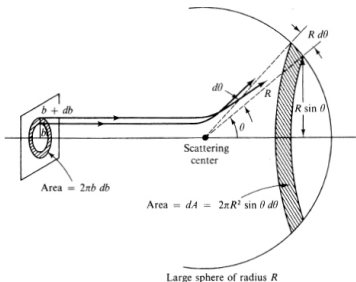


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin \theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin \theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin \theta d\theta} = \frac{b}{\sin \theta (d\theta/db)} = \frac{b}{\sin \theta |d\chi/db|} = I(\theta)$
- $\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = 2\pi \int_0^\pi I(\theta) \sin \theta d\theta$

Experimental observables: angular distributions

Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin\theta d\theta d\phi$,
where $\theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

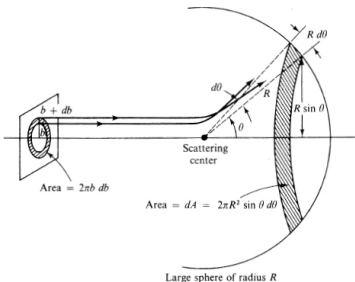


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin\theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin\theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin\theta d\theta} = \frac{b}{\sin\theta (d\theta/db)} = \frac{b}{\sin\theta |d\chi/db|} = I(\theta)$
- $\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = 2\pi \int_0^\pi I(\theta) \sin\theta d\theta$
- For realistic potentials $I(\theta)$ will be anisotropic and energy dependent.

Experimental observables: angular distributions

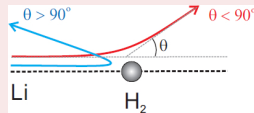
Differential cross section DCS $\frac{d\sigma}{d\Omega}$

Probability of scattering into a given solid angle $d\Omega = \sin\theta d\theta d\phi$, where $\theta < \theta < \theta + d\theta$ and $0 < \phi < 2\pi$.

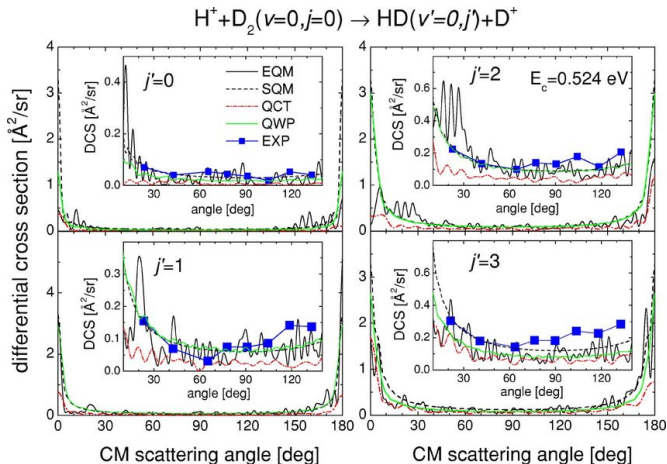


- **central potential** $\Rightarrow$ No ϕ dependence
- $d\Omega = \sin\theta d\theta \int_0^{2\pi} d\phi = 2\pi \sin\theta d\theta$
- $d\sigma = 2\pi b db$
- $\frac{d\sigma}{d\Omega} = \frac{b db}{\sin\theta d\theta} = \frac{b}{\sin\theta (d\theta/db)} = \frac{b}{\sin\theta |d\chi/db|} = I(\theta)$
- $\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = 2\pi \int_0^\pi I(\theta) \sin\theta d\theta$
- For realistic potentials $I(\theta)$ will be anisotropic and energy dependent.

DCS gives as a great deal of information on reaction mechanism



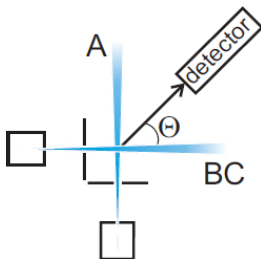
Experimental observables: angular distributions



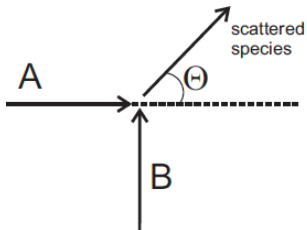
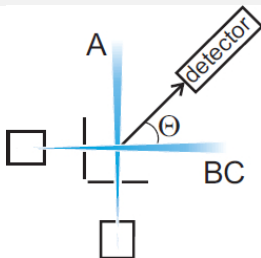
1

¹Reference: O. Roncero et al., *JCP* 128 (2008) 014304, On the dynamics of the $H^+ + D_2(v=0, j=0) \rightarrow HD + D^+$

Experimental observables: angular distributions

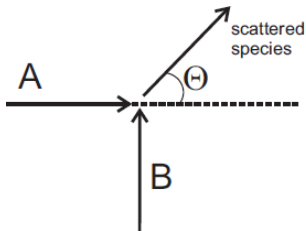
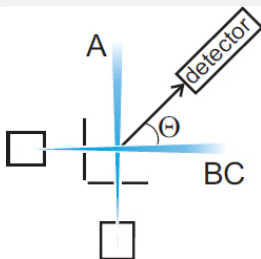


Experimental observables: angular distributions

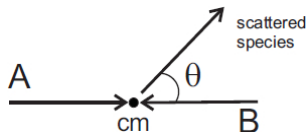


**Lab reference
frame**

Experimental observables: angular distributions

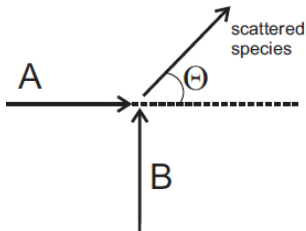
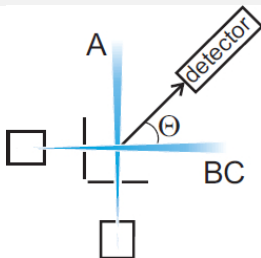


**Lab reference
frame**



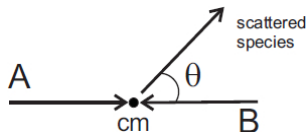
**c.o.m reference
frame**

Experimental observables: angular distributions



Lab reference
frame

change of reference frame
 $\longleftrightarrow$



c.o.m reference
frame

Reaction probability: Opacity function

Opacity function $P(b)$

$$P(b) = \frac{N_r(b_i)}{N(b_i)}$$

Fraction of collisions with b that lead to reaction

$$0 \leq P(b) \leq 1$$

- high b : centrifugal barrier $\rightarrow P(b) = 0$

Reaction probability: Opacity function

Reaction cross section $\sigma_r < \sigma$ Collision cross section

$$d\sigma_r = 2\pi b P(b) db \quad ; \quad d\sigma = 2\pi b db$$

Reaction probability: Opacity function

Reaction cross section σ_r $\sigma_r < \sigma$ Collision cross section

$$d\sigma_r = 2\pi b P(b) db \quad ; \quad d\sigma = 2\pi b db$$

$$d\sigma_{nr} = 2\pi b [1 - P(b)] db \quad (\text{non-reactive cross section})$$

Reaction probability: Opacity function

Reaction cross section $\boxed{\sigma_r < \sigma}$ Collision cross section

$$d\sigma_r = 2\pi b P(b)db \quad ; \quad d\sigma = 2\pi bdb$$

$$d\sigma_{nr} = 2\pi b[1 - P(b)]db \quad (\text{non-reactive cross section})$$

$$\sigma_r = \int_0^\infty 2\pi b P(b)db \quad ; \quad \sigma = \int 2\pi bdb$$

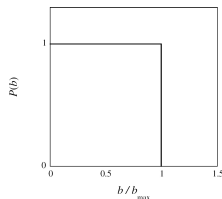
Reaction probability: Opacity function

Reaction cross section σ_r $\sigma_r < \sigma$ Collision cross section

$$d\sigma_r = 2\pi b P(b)db \quad ; \quad d\sigma = 2\pi b db$$

$$d\sigma_{nr} = 2\pi b [1 - P(b)]db \quad (\text{non-reactive cross section})$$

$$\sigma_r = \int_0^\infty 2\pi b P(b)db \quad ; \quad \sigma = \int 2\pi b db$$



Simplest approximation:

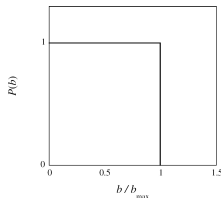
Reaction probability: Opacity function

Reaction cross section σ_r $\sigma_r < \sigma$ Collision cross section

$$d\sigma_r = 2\pi b P(b) db \quad ; \quad d\sigma = 2\pi b db$$

$$d\sigma_{nr} = 2\pi b [1 - P(b)] db \quad (\text{non-reactive cross section})$$

$$\sigma_r = \int_0^\infty 2\pi b P(b) db \quad ; \quad \sigma = \int 2\pi b db$$



Simplest approximation:

- $b \leq b_{\max} \Rightarrow P(b) = 1$
- $b > b_{\max} \Rightarrow P(b) = 0$

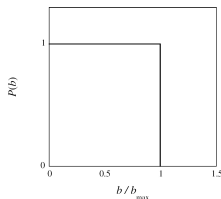
Reaction probability: Opacity function

Reaction cross section $\boxed{\sigma_r < \sigma}$ Collision cross section

$$d\sigma_r = 2\pi b P(b)db \quad ; \quad d\sigma = 2\pi bdb$$

$$d\sigma_{nr} = 2\pi b[1 - P(b)]db \quad (\text{non-reactive cross section})$$

$$\sigma_r = \int_0^\infty 2\pi b P(b)db \quad ; \quad \sigma = \int 2\pi bdb$$



Simplest approximation:

- $b \leq b_{max} \Rightarrow P(b) = 1$
- $b > b_{max} \Rightarrow P(b) = 0$

$$\hookrightarrow \sigma_r = \int_0^{b_{max}} 2\pi b P(b)db = \pi b_{max}^2$$

Reaction cross section

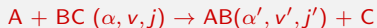


Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

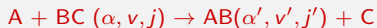
Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

Total differential cross section:
$$\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$$

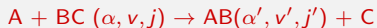
Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

Total differential cross section: $\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$

Reaction cross section

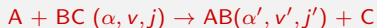


State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

Total differential cross section: $\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$

Diff. average cross section: $\frac{d\sigma_r}{d\Omega} = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$
 $= \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \frac{d\sigma_r(\alpha, \nu, j)}{d\Omega}$

Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

Total differential cross section: $\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$

Diff. average cross section: $\frac{d\sigma_r}{d\Omega} = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$
 $= \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \frac{d\sigma_r(\alpha, \nu, j)}{d\Omega}$

State-to-state integral cross section

$$\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j') = \int_0^{2\pi} \int_0^\pi \sin\theta d\theta d\phi \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$$

Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

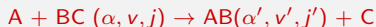
Total differential cross section: $\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$

Diff. average cross section: $\frac{d\sigma_r}{d\Omega} = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$
 $= \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \frac{d\sigma_r(\alpha, \nu, j)}{d\Omega}$

State-to-state integral cross section

$$\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j') = \int_0^{2\pi} \int_0^\pi \sin\theta d\theta d\phi \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$$

Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

Total differential cross section: $\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$

Diff. average cross section: $\frac{d\sigma_r}{d\Omega} = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$
 $= \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \frac{d\sigma_r(\alpha, \nu, j)}{d\Omega}$

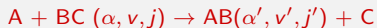
State-to-state integral cross section

$$\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j') = \int_0^{2\pi} \int_0^\pi \sin\theta d\theta d\phi \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$$

Average integral cross section

$$\sigma_r = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')$$

Reaction cross section



State-to-state differential cross section: $d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')/d\Omega$

Total differential cross section: $\frac{d\sigma_r(\alpha, \nu, j)}{d\Omega} = \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$

Diff. average cross section: $\frac{d\sigma_r}{d\Omega} = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$
 $= \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \frac{d\sigma_r(\alpha, \nu, j)}{d\Omega}$

State-to-state integral cross section

$$\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j') = \int_0^{2\pi} \int_0^\pi \sin\theta d\theta d\phi \frac{d\sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')}{d\Omega}$$

Average integral cross section

$$\sigma_r = \sum_{\alpha, \nu, j} p(\alpha, \nu, j) \sum_{\alpha', \nu', j'} \sigma_r(\alpha, \nu, j \rightarrow \alpha', \nu', j')$$

Contents

- 1 Reaction Dynamics: Introductory concepts
- 2 Collision Theory
- 3 Dynamics of elastic and reactive collisions
- 4 Theoretical Methods in Collision Dynamics**
- 5 Mechanism of Reactive Collisions
- 6 Potential Energy Surfaces

Methods for the study of Collisions Dynamics

Methods for the study of Collisions Dynamics

Theoretical methods

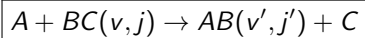
- Quasi-classic trajectory method (QCT)
- Quantum methods
 - ① Time-independent methods
 - ② Time-dependent methods

QCT Method: Introduction

QCT Method: Introduction

- QCT method assumes that the nuclei involved in a chemical reaction move according to the laws of Classical Mechanics on the potential energy surface of the system.
- "Quasiclassical" $\Rightarrow$ molecules are selected before the collision at discrete internal energy states corresponding to quantum states.
- After the collision, a "quantization" of the internal energy is also enforced on the product molecule.
- During the collision, coordinates and momenta of all atoms are calculated at any time $\Rightarrow$ animated representation of the process.
- QCT formulation can be easily generalized to systems involving a large number of atoms. (MCTDH method)
- A complete treatment of molecular collisions should include quantum effects, but in many cases QCT method leads to sufficiently accurate estimates of the experimental observables provided.

Quasi-classical dynamics



Classic hamiltonian:

$$H = T + V$$

Quasi-classical trajectory method (QCT)

Temporal evolution of the system: **Hamilton's equations**

- 6N Hamilton's equations: for three particles (N=3) there are 18 eq.

$$\begin{aligned}\dot{q}_i &= \frac{\partial H}{\partial p_i} = \frac{dq_i}{dt} \\ \dot{p}_i &= -\frac{\partial H}{\partial q_i} = \frac{dp_i}{dt}\end{aligned}$$

where $i = 1, \dots, 3N$.

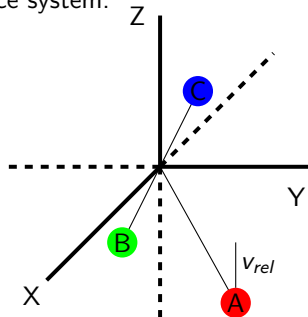
- Separate the system translation (CM) 6N-6 equations: 12 eq. (N=3)
- Total energy and total angular momentum are constant of the motion. Conservation of these quantities can serve as an **accuracy criterion** when integrating equations **numerically**.

Quasi-classical trajectory method (QCT)

Reference system:

Quasi-classical trajectory method (QCT)

Reference system:



- origin at the c.d.m. of BC
- incident atom A : plane yz
- value z of A : negative

Quasi-classical trajectory method (QCT)

Steps of a QCT calculation

Quasi-classical trajectory method (QCT)

Steps of a QCT calculation

- Setup of the initial conditions of the problem

Quasi-classical trajectory method (QCT)

Steps of a QCT calculation

- Setup of the initial conditions of the problem
- Calculation of the trajectory

Quasi-classical trajectory method (QCT)

Steps of a QCT calculation

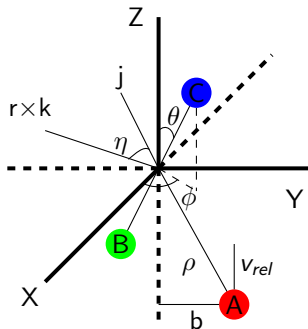
- Setup of the initial conditions of the problem
- Calculation of the trajectory
- Analysis of the products

Quasi-classical trajectory method (QCT)

Steps of a QCT calculation

- Setup of the initial conditions of the problem
- Calculation of the trajectory
- Analysis of the products
- Determination of the reaction properties

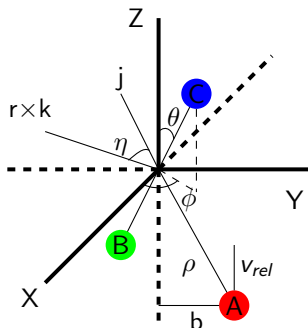
Quasi-classical trajectory method (QCT)



1.- Initial conditions of the problem

- Collision energy
- Ro-vibrational states of molecules
- Initial value of coordinates and moments

Quasi-classical trajectory method (QCT)



1.- Initial conditions of the problem

- Collision energy
- Ro-vibrational states of molecules
- Initial value of coordinates and moments

⇒ **Collision parameters:**

Geometric parameters which characterize a collision and make one collision differ from another

- 1 Distance ρ .
- 2 Impact parameter b .
- 3 Polar angles θ ($0-\pi$) y ϕ ($0-2\pi$).
- 4 Angle η ($0-\pi$).
- 5 Vibrational phase ξ ($0-\pi$).

Quasi-classical trajectory method (QCT)

Choosing the collision parameters

Quasi-classical trajectory method (QCT)

Choosing the collision parameters

- Distances ρ , ρ' : Not considerable interactions.

Quasi-classical trajectory method (QCT)

Choosing the collision parameters

- Distances ρ , ρ' : Not considerable interactions.
- b , θ , ϕ , η , ξ : Random method.

Quasi-classical trajectory method (QCT)

Choosing the collision parameters

- Distances ρ, ρ' : Not considerable interactions.
- $b, \theta, \phi, \eta, \xi$: Random method.
Random numbers are chosen ($0 \leq k_1, k_2, k_3, k_4, k_5 \leq 1$), with

$$b = b_{\max} k_1^{1/2} \quad \theta = \cos^{-1}(1 - 2k_2) \quad \phi = 2\pi k_3 \quad \eta = \pi k_4 \quad \xi = \pi k_5$$

Quasi-classical trajectory method (QCT)

2.- Calculation of a trajectory

Quasi-classical trajectory method (QCT)

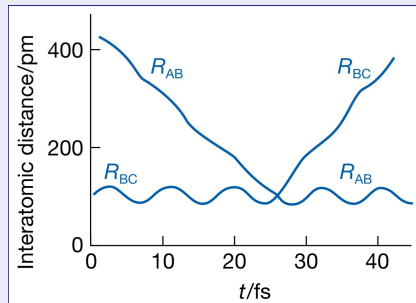
2.- Calculation of a trajectory

- Numerical integration of the Hamilton's equations
- Methods:
 - ▶ Runge-Kutta method
 - ▶ Adams - Moulton method
- **Optimize integration step:** check energy and angular momentum conservation
- Another check: run the trajectory in the reverse direction

Quasi-classical trajectory method (QCT)

2.- Calculation of a trajectory

- Numerical integration of the Hamilton's equations
- Methods:
 - ▶ Runge-Kutta method
 - ▶ Adams - Moulton method
- **Optimize integration step:** check energy and angular momentum conservation
- Another check: run the trajectory in the reverse direction



Quasi-classical trajectory method (QCT)

3.- Analysis of products: $A + BC \rightarrow AB + C$

- Determination of: **Check if the energy is conserved**
 - ▶ Relative translational, vibrational and rotational energy
 - ▶ Rotational angular momentum of the final molecule
 - ▶ *Scattering* angle
- Assignment of final quantum numbers:

$$j' \leftarrow \tilde{j}' = -\frac{1}{2} + \frac{1}{2} \sqrt{1 + \frac{4\mathbf{j}_{AB}\mathbf{j}_{AB}}{\hbar^2}}$$

$$v' \leftarrow \tilde{v}' = -\frac{1}{2} + \frac{1}{\pi\hbar} \int_{r-}^{r+} \sqrt{2\mu_{AB} \left(E_{int}(AB) - V(r_{AB}) - \frac{\mathbf{j}_{AB}\mathbf{j}_{AB}}{2\mu_{AB}r_{AB}^2} \right)}$$

Quasi-classical trajectory method (QCT)

Steps of the QCT method

- 1 Choose the random numbers (k_1, k_2, k_3, k_4, k_5)
- 2 Determine the geometric collision parameters ($b, \theta, \phi, \eta, \xi$)
- 3 Establish the initial values of the Jacobi's coordinates and the moments
- 4 Integrate the equations of motion
- 5 Identify the products and determine the energetic distribution
- 6 Evaluate the magnitudes that characterize the reactivity

The outcome of a single trajectory is insufficient to describe the reaction characteristics. A statistically significant sample of trajectories has to be integrated and related outcomes averaged

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- Opacity function:

$$P_r(v_r; b_i) = \frac{N_r(v_r; b_i)}{N(v_r; b_i)}$$

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- Opacity function:

$$P_r(v_r; b_i) = \frac{N_r(v_r; b_i)}{N(v_r; b_i)}$$

- Reactive cross section:

$$\sigma_{vj}(v_r) = \pi b_{max}^2 \frac{N_r(v_r; v, j)}{N}$$

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- Opacity function:

$$P_r(v_r; b_i) = \frac{N_r(v_r; b_i)}{N(v_r; b_i)}$$

- Reactive cross section:

$$\sigma_{vj}(v_r) = \pi b_{max}^2 \frac{N_r(v_r; v, j)}{N}$$

- State-to-state reactive cross section:

- ▶ Integral:

$$\sigma_{v',j' \leftarrow v,j}(v_r) = \pi b_{max}^2 \frac{N_{v',j';v,j,v',j'}}{N}$$

- ▶ Differential:

$$\frac{d\sigma_{v',j' \leftarrow v,j}(v_r)}{d\omega}(\theta) = \frac{\sigma_{v',j' \leftarrow v,j}(v_r)}{2\pi} \sum_{k=0}^{\infty} \frac{2k+1}{2} s_k P_k(\cos \theta)$$

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- State-to-state rate coefficients with an average over the translational energies

$$K(T_{tr}; v, j, v', j') = \sqrt{\frac{8k_B T_{tr}}{\pi \mu_{A,BC}}} \pi b_{max}^2 \frac{N_r(v_r(T_{tr}); v, j, v', j')}{N(v_r(T_{tr}); v, j, v', j')}$$

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- State-to-state rate coefficients with an average over the translational energies

$$K(T_{tr}; v, j, v', j') = \sqrt{\frac{8k_B T_{tr}}{\pi \mu_{A,BC}}} \pi b_{max}^2 \frac{N_r(v_r(T_{tr}); v, j, v', j')}{N(v_r(T_{tr}); v, j, v', j')}$$

- Rotational-averaging rate constant:

$$K(T_{tr}, T_{rot}; v, v', j') = \sum_j P_j(T_{rot}) K(T_{tr}; v, j, v', j')$$

where

$$P_j(T_{rot}) = \frac{(2j+1)e^{E_{vj}/k_B T_{rot}}}{\sum_j (2j+1)e^{E_{vj}/k_B T_{rot}}}$$

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- Rotational-averaging rate constant:

$$K(T_{tr}, T_{rot}; \nu) = \sum_{\nu', j'} K(T_{tr}, T_{rot}; \nu, \nu', j')$$

Quasi-classical trajectory method (QCT)

Determination of the reaction properties

- Rotational-averaging rate constant:

$$K(T_{tr}, T_{rot}; \nu) = \sum_{\nu', j'} K(T_{tr}, T_{rot}; \nu, \nu', j')$$

- thermal averaging rate coefficients (to compare with kinetic experiments):

$$K(T) = \sum_{\nu} P_{\nu}(T) K(T_{tr}, T_{rot}; \nu)$$

where

$$P_{\nu}(T) = \frac{e^{E_{\nu}/k_B T}}{\sum_{\nu} e^{E_{\nu}/k_B T}}$$

Quasi-classical trajectory method (QCT)

Angular distribution of the products

- Obtaining the reactive differential cross section (DCS)

→ Comparison to the experimental DCS .

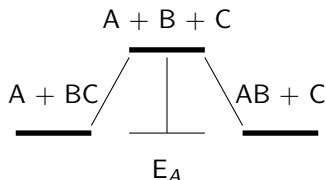
- Scattering angles:
 - ▶ Small: collision *forward*
 - ▶ Big: collision *backward*
 - ▶ $\theta \approx 90^\circ$: collision *sideways*

Contents

- 1 Reaction Dynamics: Introductory concepts
- 2 Collision Theory
- 3 Dynamics of elastic and reactive collisions
- 4 Theoretical Methods in Collision Dynamics
- 5 Mechanism of Reactive Collisions**
- 6 Potential Energy Surfaces

Study of the collisions $A + BC \rightarrow AB + C$

Is it sensible to think that, when a chemical reaction takes place, the new bonds are formed only after the old ones have been broken?



In order for this to happen, we would need to have enough energy to break a bond. Those energies are usually very high ($> 200 \text{ kJ/mol}$), what implies “activation energies” large enough as to render the reaction velocity extraordinarily small.

Mechanism of Reactive Collisions

Types of reactive collisions

- 1 Direct-type collisions $\rightarrow t < 10^{-14} \text{ s}$
- 2 Compound collisions: formation of long-lived complexes $\rightarrow t \geq 10^{-12} \text{ s}$

Mechanism of Reactive Collisions

Types of reactive collisions

- 1 Direct-type collisions $\rightarrow t < 10^{-14} \text{ s}$
- 2 Compound collisions: formation of long-lived complexes $\rightarrow t \geq 10^{-12} \text{ s}$

The angular distribution as a probe of direct versus compound collisions

How is the mechanism shown in the angular distributions?

Mechanism of Reactive Collisions

Types of reactive collisions

- 1 Direct-type collisions $\rightarrow t < 10^{-14} \text{ s}$
- 2 Compound collisions: formation of long-lived complexes $\rightarrow t \geq 10^{-12} \text{ s}$

The angular distribution as a probe of direct versus compound collisions

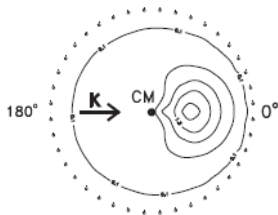
How is the mechanism shown in the angular distributions?

Direct reactions $\rightarrow$ asymmetric angular distributions

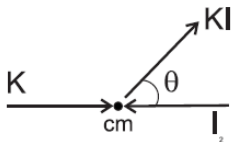
Compound reactions $\rightarrow$ Forward-backward symmetric angular distributions

Direct-type reaction: Harpoon mechanisms

Example: $\text{K} + \text{I}_2 \rightarrow \text{KI} + \text{I}$



- Plot data in the form of a contour plot in the c.o.m system
- Contours represent lines of equal product flux as a function of KI velocity (i.e. flux as a function of scattering angle (θ) and KI speed)
- The outer circle represents the maximum product velocity, so less vibrational energy (internal states of products).
- Scattering angles $\rightarrow 0$: forward scattering
- Scattering angles $\rightarrow 180$: backward scattering



- Observed forward scattering ($\theta < 90^\circ$)

Direct-type reaction: Harpoon mechanisms

Observations:

- very large cross sections (like many other alkali + halogen reactions)
- forward scattering products
- low energy going into translation. KI highly internally excited

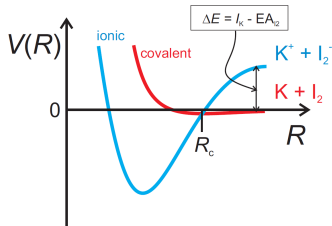
Direct-type reaction: Harpoon mechanisms

Observations:

- very large cross sections (like many other alkali + halogen reactions)
- forward scattering products
- low energy going into translation. KI highly internally excited

The harpoon mechanism:

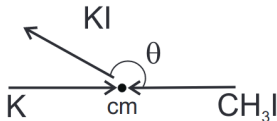
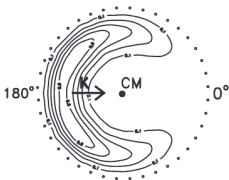
- proposed by Michael Polanyi (1932) to explain the very large σ_r of alkali atoms with halogen molecules
- it is a long-range electron transfer process (at large distances).
- it applies for reactions of atoms with low ionization potentials (I) with molecules with high electron affinity (EA)



$$I_K - EA_{I_2} = \frac{e^2}{4\pi\epsilon_0 R_c}$$
$$\hookrightarrow R_c \approx 6 \text{ \AA}$$

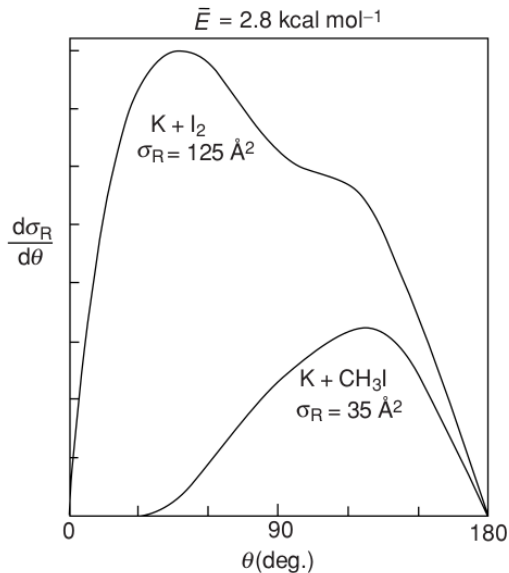
K^+ carries I^- in its original direction (forward scattering)

Direct-type reaction: Rebound or backward dynamics



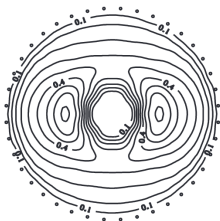
- small cross sections
- backward scattering products (consistent with small values of b)
- high translational energy of the products
- it is also a harpoon mechanism but in case the crossing occurs at short range (small b) $\rightarrow$ backward scattering

Direct-type reaction: angular distribution

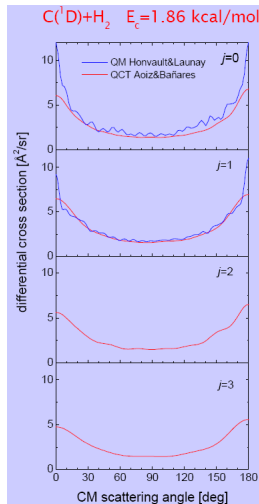


Reactions with long life complex formation

Forward-backward symmetric angular distributions

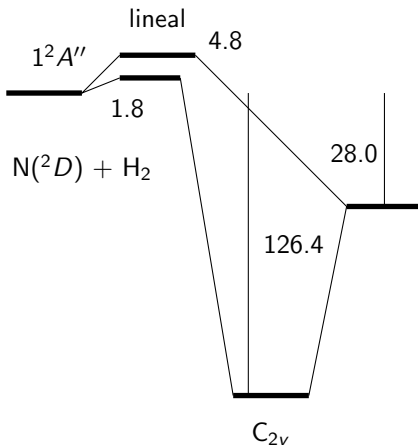
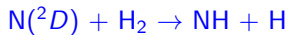


- Complex lifetime must be long compared to the rotational period
- Favoured by reactions on attractive potential energy surfaces



Reactions with long life complex formation

Insertion reaction



Contents

- 1 Reaction Dynamics: Introductory concepts
- 2 Collision Theory
- 3 Dynamics of elastic and reactive collisions
- 4 Theoretical Methods in Collision Dynamics
- 5 Mechanism of Reactive Collisions
- 6 Potential Energy Surfaces**

Potential energy surface

Potential energy surface

Born-Oppenheimer Approximation

$$\hat{H}\Psi(\mathbf{r}, \mathbf{R}) = E_{tot}\Psi(\mathbf{r}, \mathbf{R})$$

Potential energy surface

Born-Oppenheimer Approximation

$$\hat{H}\Psi(\mathbf{r}, \mathbf{R}) = E_{\text{tot}}\Psi(\mathbf{r}, \mathbf{R})$$

Electronic Schrödinger equation:

$$\left(\hat{H}_{\text{el}} + V_{\text{NN}}\right)\psi_{\text{el}}(\mathbf{r}; \mathbf{R}) = V(\mathbf{R})\psi_{\text{el}}(\mathbf{r}; \mathbf{R})$$

where $V(\mathbf{R}) = E_{\text{el}} + E_{\text{NN}}$.

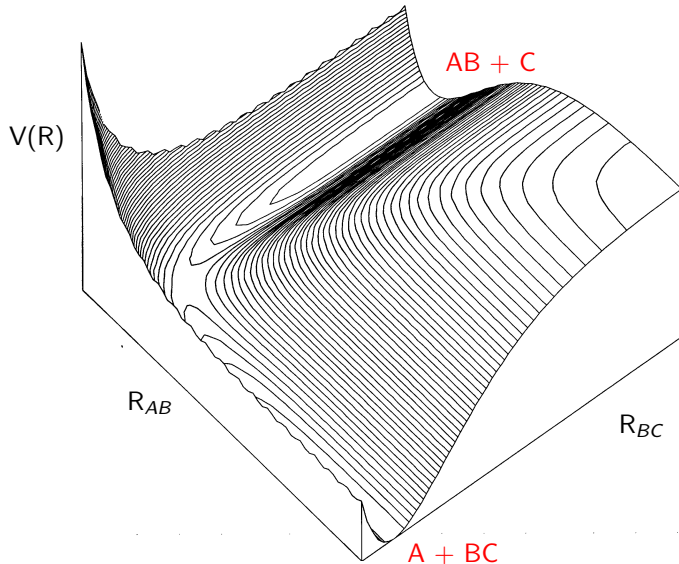
Nuclear Schrödinger equation:

$$\hat{H}_{\text{nuc}}\psi_{\text{nuc}}(\mathbf{R}) = \left(\hat{T}_{\text{nuc}}(\mathbf{R}) + V(\mathbf{R})\right)\psi_{\text{nuc}}(\mathbf{R}) = E_{\text{tot}}\psi_{\text{nuc}}(\mathbf{R})$$

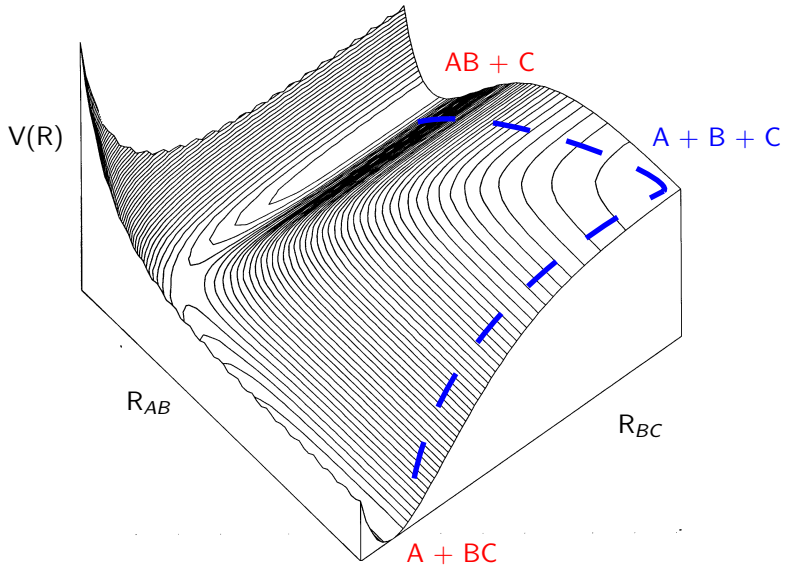
$V(\mathbf{R})$: potential energy for the nuclei motion. It corresponds to the field created by the electrons.

$\psi_{\text{nuc}}(\mathbf{R})$: nuclei motion (rotation, vibration and system displacement; scattering properties).

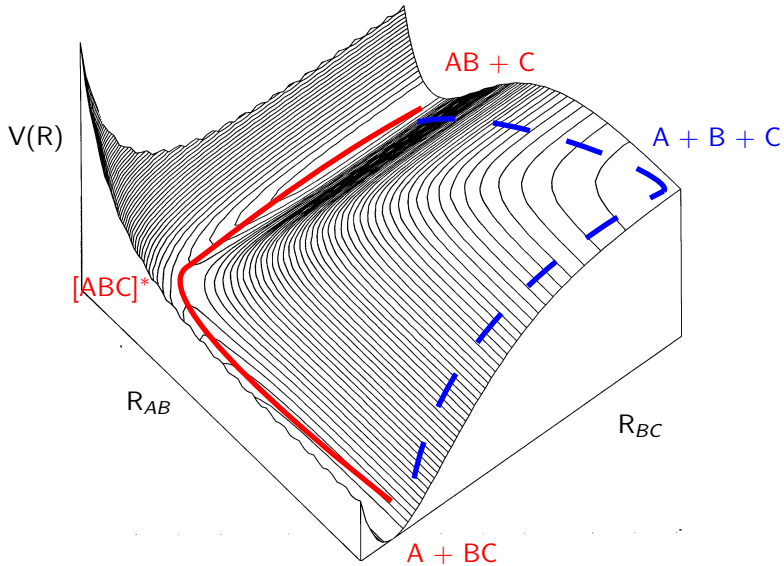
Potential energy surface



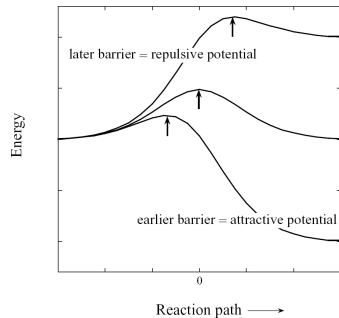
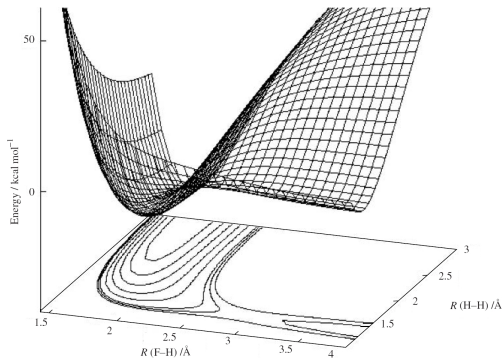
Potential energy surface



Potential energy surface



Motion at the PES. Reaction path



Motion at the PES. Reaction path

Polanyi's rules

Consider the motion of the nuclei on the PES to be classical

The location of the transition states is important



- Early barrier

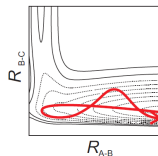
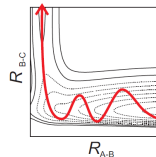
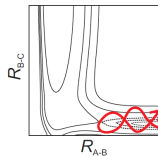
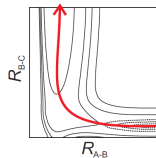
The kinetic energy will be effective (left)

Vibrational energy will be ineffective (right)

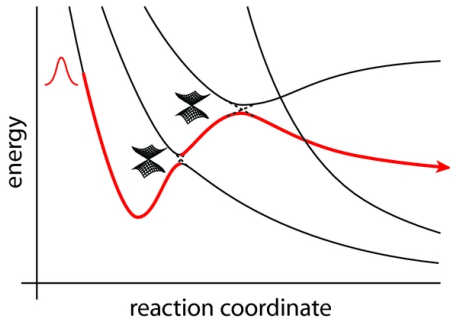
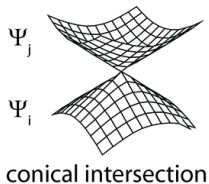
- Late barrier

The kinetic energy will be effective (left)

The kinetic energy will be effective (right)



Diabatic and non-adiabatic processes



- Born-Oppenheimer approximation no longer valid
- Calculation of non-adiabatic couplings
- **Surface Hopping Calculations** (SH-QCT)